

SAAD MUHAMMAD KHAN

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EDUCATION

National University of Science & Technology (NUST)

Expected September 2028

Bachelor of Computer Science

EXPERIENCE

Independent Software Developer

- **Python Development (ML, Data Science)** — Built **deep learning pipelines** using **CNNs** and **BERT-based transformer models**; implemented data preprocessing, normalization and feature engineering
- **Full-Stack Development** — Developed **React.js** web applications and **RESTful APIs** using **Node.js** and **Express.js**; designed database schemas, data persistence layers, and query-based retrieval systems
- **Mobile App Development** — Built mobile applications using **Flutter** and **Java** with backend API integration

Java Programming Virtual Internship | CodeAlpha

- Completed a virtual internship focused on **OOP**, **problem-solving**, and Java application development.

Summer Internship | Alkhidmat Foundation Pakistan

- Completed a **6-week internship** in the **Volunteer Management Department**.

PROJECTS

Image Mosaic to Hyperspectral Imaging (HSI) Reconstruction | Python | PyTorch | CNNs

- Developed a **deep learning pipeline** to reconstruct **61-band hyperspectral cubes** from low-cost mosaic inputs for the **Hyper-Object Challenge**
- Implemented **CNN-based spectral reconstruction models** and preprocessing pipelines for **normalization**, **tensor reshaping**, and **spectral band alignment**
- Processed and handled **1024×1024×61 hyperspectral datasets** for multi-band spectral cube generation

BERT-Based Multilingual Polarization Detection Model | Python | PyTorch | Transformers | NLP

- Built a **multilingual text polarization detection system** across English, Chinese, Urdu, Arabic, and Turkish datasets
- Fine-tuned **BERT-based transformer models** including **RoBERTa-Large** and **XLNet** using **PyTorch** and **Hugging Face Transformers** with multilingual tokenization and subword encoding
- Applied **class-weighted cross-entropy loss** for imbalanced datasets and uploaded trained models to **Hugging Face Hub** for deployment and accessibility

Bankchain | Modular Blockchain Banking System | Solidity, Hardhat, Ethers.js, React

- Developed a **modular Ethereum-based banking system** using **Solidity**, **Hardhat**, and **OpenZeppelin**, implementing a **multi-contract architecture** for deposits, withdrawals, transfers, loan lifecycle, and card management
- Integrated **Ethers.js** with **React** for wallet-based interaction and real-time smart contract communication, and implemented **on-chain audit logging**, security controls (**Ownable**, **ReentrancyGuard**), and a full **Hardhat workflow** for deployment, testing, and execution

WardEase | Ward Management System (Android Application) | Java, Firebase

- Developed **WardEase**, an Android hospital ward management system using **Java** and **Firebase Realtime Database** with real-time patient data synchronization at **<2s latency**, supporting **50+ concurrent users**
- Implemented modules for **patient records**, **ward allocation**, and **status tracking**, reducing manual record handling by **~40–50%**
- Designed a **role-based access control system** (admin/doctor/staff), reducing unauthorized modifications by **~60%** and replacing manual logs with a **centralized digital system**

Sports Analytics Website (Web Application) | MERN Stack

- Built a full-stack sports analytics platform using **React**, **Node.js**, **Express.js**, and **MongoDB** for multiple sports
- Designed **RESTful APIs** and **MongoDB schemas** for match statistics and player metrics (**17,000+** sample records), tested across **10+ API routes** using **Postman**
- Developed a **React-based analytics dashboard** for interactive visualization of sports performance data and trends

TECH STACK

Frontend Development: HTML5, CSS3, JavaScript (ES6+), React.js,

Backend Development: Node.js, Express.js, Flask, JDBC, RESTful API Development, Authentication Systems

Machine Learning: Python (NumPy, Pandas, Scikit-learn), PyTorch, TensorFlow, Transformers, NLP, CNN

Mobile Development: Flutter, Java, Responsive UI/UX Design

Databases: PostgreSQL, MySQL, Oracle Database, MongoDB, Firebase, Redis, Neo4j

DevOps & Tools: Docker, Git, GitHub, Postman, VS Code

Programming Languages: Python, JavaScript, C++, Java, Dart, Assembly Language (AVR, MASM)

ACADEMIC PERFORMANCE

Maintained consistent academic excellence with a CGPA of **3.79 / 4.00**, demonstrating strong performance in core computer science subjects and analytical problem-solving skills.